QP CODE: 25901145

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBBA200 - STATISTICS FOR BUSINESS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

SCHEME OF VALUATION

Part A

Short Answer Questions. Answer any **5** questions

Give **3** marks to each correct answer

1. Definition -3 marks
2. One limitation-3 marks
3. Definition-3 marks
4. One merit-1.5 marks, One demerit- 1.5 marks
5. Paired t test- 1.5 marks, Independent t test- 1.5 marks
6. Definition – 3 marks

Part B

Short Essay Questions. Answer any 3 questions

Give 5 marks to each correct answer

7. Actual data=T+S+C+I – 2 Marks, S=Actual data-Trend – 1 Mark
   Correct calculation – 2 Marks
8. Four Year moving average- 2.5 marks, Five year moving average-2.5 marks
9. Two applications of t test- 2.5 marks each
10. Write the null and alternate hypotheses, Formula for Kruskal Wallis test statistic, Decision rule- 5 marks

Part C

Essay Questions. Answer any 2 questions

Give 10 marks to each correct answer

11. Any reasonable attempt – 10 marks
12. Trend line by Free hand curve method- 10 marks
13. Null Hypothesis: There is no significant difference between the means,
    Alternate Hypothesis: At least one of the means is not equal - 1 mark
    $T=15+18+21=54$, $N=3+3+3=9$, Correction Factor=$T^2/N= 324$ - 2 marks
    Total Sum of Squares=12 - 1 mark
    Sum of Squares between Columns=6 - 1 mark
    Error Sum of Squares=12-6=6 - 1 mark
    Mean Sum of Squares(C)=6/2=3 - 0.5 mark
    Mean Sum of Squares(E)=6/6=1 - 0.5 mark
    F statistic=3 - 1 mark

$F_{(2,6)}$ at 5% level of significance=4.26 - 1 mark

Calculated value of F <Tabled value of F. Therefore, Accept Null Hypothesis.
- 1 mark

10 marks for correct answer

14. Null Hypothesis: Preference for type of music and age are independent,
Alternate Hypothesis: Preference for type of music and age are dependent - 1 mark

The Expected frequencies are 70.24, 176.82, 58.94, 98.7, 248.47, 82.82, 61.06, 153.27, 51.24 - 3 marks

Chi-square statistic=314.18 - 4 marks

$\varkappa^2_{(4)}$ at 5% level of significance=9.488 - 1 mark

Calculated value of Chi square >Tabled value of Chi square. Therefore, Reject Null Hypothesis. - 1 mark

10 marks for correct answer